Job Description (JD)

Position Title: Design Validation & Testing Engineer – EV Power Electronics

Department: Research & Development (R&D)

Industry: Electric Vehicles / Automotive Electronics

JOB SUMMARY

We are looking for a Design Validation & Testing Engineer to support the evaluation, testing, and validation of power electronics and electronic control systems used in electric vehicles and energy products. The role involves hands-on testing, system validation, compliance verification, and failure analysis to ensure products meet performance, safety, reliability, and regulatory standards from prototype to mass production.

KEY RESPONSIBILITIES

• Execute design validation and testing activities across EVT, DVT, and PVT phases

• Develop and perform functional, performance, endurance, and stress tests for hardware and embedded systems

• Set up and operate electrical, thermal, and mechanical test benches

• Perform performance, efficiency, and reliability testing and analysis

• Validate system behavior at vehicle level, including start-up, shutdown, and fault conditions

• Ensure compliance with automotive and international standards (AIS, IS, IEC, ISO, SAE, UL, CE)

• Support EMI/EMC, electrical safety, insulation, and leakage current testing

• Conduct failure analysis and support root cause investigations (RCA)

• Prepare validation reports, test documentation, and release readiness data

• Coordinate with design, firmware, manufacturing, quality, and supplier teams

TECHNICAL SKILLS

• Knowledge of power electronics fundamentals and converter topologies

• Hands-on experience with lab equipment such as oscilloscopes, power analyzers, electronic loads, and HV probes

• Familiarity with simulation tools (MATLAB/Simulink, LTSpice, PLECS)

• Understanding of EMI/EMC and electrical safety testing

• Working knowledge of embedded systems and communication protocols (CAN, Ethernet)

• Basic experience with Python or scripting for test automation and data analysis

• Ability to read and review schematics and PCB layouts

SOFT SKILLS

• Strong analytical and problem-solving skills

• Attention to detail and quality orientation

• Good communication and documentation skills

• Ability to work independently and in cross-functional teams

• Willingness to learn and adapt in a fast-paced environment

WHAT WE OFFER

• Opportunity to work on advanced EV and power electronics technologies

• Exposure to the complete product lifecycle from development to production

• Collaborative and growth-oriented work environment